9. Playfair Decryption – PT-109 Message

#include <stdio.h> #include <string.h> #include <ctype.h>

char matrix[5][5] = { {'R','O','Y','A','L'}, {'N','E','W','Z','D'}, {'V','B','C','F','G'}, {'H','I','K','M','P'}, {'Q','S','T','U','X'} };

void findPosition(char ch, int *row, int *col) { int i, j;

```
if (ch == 'J')
    ch = 'I';

for (i = 0; i < 5; i++)
{
    for (j = 0; j < 5; j++)
    {
        if (matrix[i][j] == ch)
        {
            *row = i;
            *col = j;
            return;
        }
    }
}


}
```

void decrypt(char a, char b) { int r1, c1, r2, c2;

```
findPosition(a, &r1, &c1);
findPosition(b, &r2, &c2);

if (r1 == r2)
{
    printf("%c%c",
           matrix[r1][(c1 + 4) % 5],
           matrix[r2][(c2 + 4) % 5]);
}
else if (c1 == c2)
```

```
{
    printf("%c%c",
           matrix[(r1 + 4) % 5][c1],
           matrix[(r2 + 4) % 5][c2]);
}
else
{
    printf("%c%c",
           matrix[r1][c2],
           matrix[r2][c1]);
}


}
```

int main() { char cipher[] = "KXJEYUREBEZWEHEWRYTUHEYFS"
"KREHEGOYFIWTTTUOLKSYCAJPO" "BOTEIZONTXBYBNTGONEYCUZWR"
"GDSONSXBOUYWRHEBAAHYUSEDQ";

```
int i;

printf("Decrypted Message:\n\n");

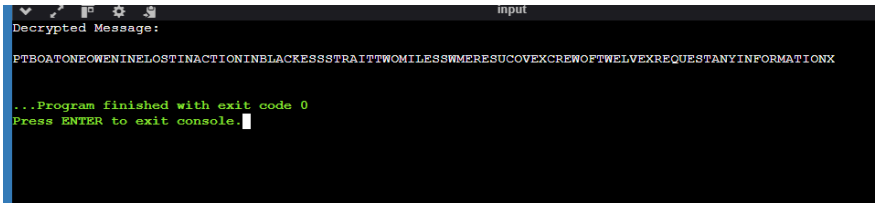
for (i = 0; i < strlen(cipher); i += 2)
{
    decrypt(cipher[i], cipher[i + 1]);
}

printf("\n");

return 0;


}
```

OUTPUT


input
Decrypted Message:
PTBOATONEOWENINELOSTINACTIONINBLACKESSSTRAITTWOMILESWMERESUCOVEXCREWOFTWELVEXREQUESTANYINFORMATIONX
...Program finished with exit code 0
Press ENTER to exit console.